A Process-Relational Presentation of Finite Metric Geometry

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Abstract

This paper develops a conceptual framework for finite discrete geometry organized around three commitments: geometric structure arises from generators and composition laws rather than from a background space; objects are equivalence classes of finite processes under a congruence; and metric structure is an external parameter rather than an intrinsic property of geometric objects.

The mathematical language used throughout is that of quotients of free categories on weighted directed graphs — structures well-established in geometric group theory, the theory of groupoids, and enriched category theory. The principal constructions (metric emergence from path optimization, universality for finite metric spaces, categorical equivalence with Lawvere metric spaces, Gromov–Hausdorff compatibility) are instances of known results, recast within this framework and collected in one place. The relationship to existing work is stated explicitly in Section 11.

The primary new mathematical result is a complete isomorphism classification of single-generator congruences by an invariant pair (n, k) of period and shift order (Proposition 4.10). The main contribution of the paper is a *presentation-level semantic framework*: geometric structure is defined at the level of generators and congruence laws, prior to quotienting, and metric properties arise as invariants of congruence classes of processes. This separates layers — syntax (generators), laws (congruence), objects (quotient), and distance (emergent metric) — that are usually conflated in classical geometry and in geometric group theory expositions.

Scope and Organization

A reader familiar with geometric group theory or enriched category theory will recognize the structures below as weighted directed graphs, path metrics, and their categorical formulation via Lawvere's identification of metric spaces with enriched categories [1]. The exposition is self-contained: all definitions are given from scratch, and no prior knowledge of category theory is assumed. The framework's relationship to existing work — including what is reformulated and what is genuinely new — is stated in Section 11.

Notation. V denotes a set of states, R a set of generators, $s, t : R \to V$ source and target maps, $\mathcal{C}$ the set of all configurations over (V, R, s, t), and $\sim$ a congruence on $\mathcal{C}$. We write $[C]$ for the equivalence class of a configuration C, and $\mathcal{C}/\!\sim$ for the quotient. The empty configuration at state v is id_v, written id when v is clear from context.

1 Relational Systems

Definition 1.1 (Relational System)**.** A *relational system* is a triple (V, R, s, t) where:

- V is a non-empty set of *states*,
- R is a non-empty set of *generators*,
- $s, t : R \to V$ are the *source* and *target* maps, assigning to each generator a domain and codomain state.

Definition 1.2 (Compatibility)**.** Generators $r_1, r_2 \in R$ are *compatible* if $t(r_1) = s(r_2)$.

Remark 1.3 (Graph structure). A relational system (V, R, s, t) is precisely a directed graph with vertex set V and edge set R. Generators are edges; compatibility is the standard condition for path concatenation; configurations are directed paths. A weight function $w : R \to [0, \infty)$ may be added when metric structure is needed (Section 5). In Section 8 this graph is the generating data for a free enriched category, and the congruence $\sim$ plays the role of relations in the categorical sense of a *presentation by generators and relations*.

2 Configurations as Finite Processes

Definition 2.1 (Configuration)**.** A *configuration* is a finite sequence of generators

$$C = (r_1, r_2, \ldots, r_n), \quad r_i \in R,$$

such that $t(r_i) = s(r_{i+1})$ for each $i = 1, \ldots, n-1$ (each consecutive pair is compatible in the sense of Definition 1.2). The *source* and *target* of C are $s(C) := s(r_1)$ and $t(C) := t(r_n)$. The *empty configuration* id_v at state v (of length 0) satisfies $s(\mathrm{id}_v) = t(\mathrm{id}_v) = v$ and acts as a two-sided identity. We write id when the state is clear from context. The set of all configurations is denoted $\mathcal{C}$.

Definition 2.2 (Configuration Composition)**.** For $C = (r_1, \ldots, r_m)$ and $D = (s_1, \ldots, s_n)$ with $t(C) = s(D)$ (i.e., $t(r_m) = s(s_1)$), their *composition* is concatenation:

$$C \circ D = (r_1, \ldots, r_m, s_1, \ldots, s_n).$$

Composition is associative and $\mathrm{id}_v \circ C = C$, $C \circ \mathrm{id}_v = C$ whenever the sources and targets match.

Remark 2.3 (Configurations are paths). A configuration is precisely a directed path in the graph (V, R, s, t) of Definition 1.1. Composition is path concatenation, which is associative by construction. No further associativity axiom is required.

3 Congruence and Observational Indistinguishability

Definition 3.1 (Congruence)**.** An equivalence relation $\sim$ on $\mathcal{C}$ is a *congruence* (with respect to composition) if it satisfies:

(i) **Typing preservation.** $C_1 \sim C_2$ implies $s(C_1) = s(C_2)$ and $t(C_1) = t(C_2)$. Congruent configurations share the same source and target state.

(ii) **Composition compatibility.** For all $C_1, C_1', C_2, C_2' \in \mathcal{C}$,

$$C_1 \sim C_1' \text{ and } C_2 \sim C_2' \;\Rightarrow\; C_1 \circ C_2 \sim C_1' \circ C_2',$$

whenever the compositions are defined.

Typing preservation ensures that the quotient composition $[C_1] \circ [C_2]$ is well-typed: compatible classes remain compatible after quotienting.

Remark 3.2 (Practical scope of congruences). The definition above allows arbitrary congruences. In practice, the paper restricts attention to congruences *generated by a finite set of relational laws* — that is, the smallest congruence containing a given set of pairs (C_i, C_i'), closed under the conditions above. This is the setting of category presentations. Additional properties — such as confluence (the Church–Rosser property) or termination of the associated rewriting system — may be imposed when decidability or algorithmic metric computation is required. Without such properties, membership in a congruence class may be undecidable and metric values may be uncomputable. The examples throughout the paper use finitely generated congruences with straightforward normal forms.

Definition 3.3 (Observational Indistinguishability)**.** Given a congruence $\sim$ on $\mathcal{C}$, configurations C_1 and C_2 are *observationally indistinguishable with respect to* $\sim$ if, for every $D \in \mathcal{C}$:

$$C_1 \circ D \sim C_2 \circ D \quad (\text{whenever } C_1, C_2 \text{ are left-compatible with } D),$$

$$D \circ C_1 \sim D \circ C_2 \quad (\text{whenever } D \text{ is left-compatible with } C_1, C_2).$$

Remark 3.4 (Derived property). Observational indistinguishability is a consequence of congruence, not a definition of it. If $C_1 \sim C_2$ and $\sim$ is a congruence, then for any compatible D: $C_1 \circ D \sim C_2 \circ D$ follows immediately from Definition 3.1 applied with $D \sim D$. The intended workflow is: (1) fix a relational system; (2) specify $\sim$ by listing relational laws; (3) verify Definition 3.1; (4) conclude that equivalent configurations are observationally indistinguishable.

Definition 3.5 (Relational Inverse)**.** Given a relational system (V, R, s, t) with congruence $\sim$, a generator $r \in R$ has an *inverse* $r^{-1} \in R$ if:

$$(r, r^{-1}) \sim \mathrm{id}_{t(r)} \quad \text{and} \quad (r^{-1}, r) \sim \mathrm{id}_{s(r)}.$$

A relational system is *invertible* (with respect to $\sim$) if every generator has an inverse. Note that r^{-1} must satisfy $s(r^{-1}) = t(r)$ and $t(r^{-1}) = s(r)$ for the configurations (r, r^{-1}) and (r^{-1}, r) to be well-formed.

4 Objects and the Quotient Construction

Definition 4.1 (Object)**.** An *object* is an equivalence class $[C]$ of a configuration C under a congruence $\sim$:

$$[C] = \{C' \in \mathcal{C} \mid C' \sim C\}.$$

The set of all objects is the quotient $\mathcal{C}/\!\sim$.

Remark 4.2 (Terminology and categorical context). In the categorical language of Section 8, the equivalence classes $[C]$ are *morphisms* in the quotient category, while the states V of the relational system serve as categorical *objects*. The term "object" is used here in the geometric sense—an equivalence class of processes that no relational test can distinguish—not in the categorical sense. The two usages are reconciled in Section 8.

Theorem 4.3 (Quotient Construction)**.** *Let (V, R, s, t) be a relational system with configuration set $\mathcal{C}$, and let $\sim$ be a congruence on $\mathcal{C}$. Then:*

(i) The quotient $\mathcal{C}/\!\sim$ is well-defined as a set of equivalence classes.

(ii) Composition descends to a well-defined partial operation on $\mathcal{C}/\!\sim$: whenever $t(C_1) = s(C_2)$,

$$[C_1] \circ [C_2] \;:=\; [C_1 \circ C_2].$$

(iii) The induced composition is associative.

(iv) For each state $v \in V$, the class $[\mathrm{id}_v]$ acts as a two-sided identity for all classes $[C]$ with $s(C) = v$ or $t(C) = v$.

Proof. **(i).** $\sim$ partitions $\mathcal{C}$ into disjoint non-empty equivalence classes.

(ii). Suppose $[C_1] = [C_1']$ and $[C_2] = [C_2']$, i.e., $C_1 \sim C_1'$ and $C_2 \sim C_2'$. Since $\sim$ is a congruence, $C_1 \circ C_2 \sim C_1' \circ C_2'$, so $[C_1 \circ C_2] = [C_1' \circ C_2']$. The operation is independent of representatives. Compatibility of $[C_1]$ and $[C_2]$ is inherited from the compatibility of their representatives.

(iii). For $C_1, C_2, C_3 \in \mathcal{C}$, associativity at the configuration level gives $(C_1 \circ C_2) \circ C_3 = C_1 \circ (C_2 \circ C_3)$. By part (ii):

$$([C_1] \circ [C_2]) \circ [C_3] = [(C_1 \circ C_2) \circ C_3] = [C_1 \circ (C_2 \circ C_3)] = [C_1] \circ ([C_2] \circ [C_3]).$$

(iv). For any C starting at v, $\mathrm{id}_v \circ C = C$, hence $[\mathrm{id}_v] \circ [C] = [C]$. Similarly on the right. $\square$

4.1 Taxonomy of Single-Generator Quotients

Proposition 4.4 (Conditions for a cyclic closed quotient)**.** *Let (V, R, s, t) be a relational system with a single generator a and congruence $\sim$. The quotient $\mathcal{C}/\sim$ is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ if and only if:*

(i) ***Closure.*** *$a^n \sim \mathrm{id}$ for some $n \geq 1$.*

(ii) ***Uniqueness.*** *$a^j \not\sim a^k$ for all $0 \leq j < k < n$.*

When closure holds but uniqueness fails, the quotient has period strictly less than n. When closure fails, all powers are distinct.

Proof. If both conditions hold, the n classes $[a^0], \ldots, [a^{n-1}]$ are distinct and $\mathcal{C}/\sim \cong \mathbb{Z}/n\mathbb{Z}$ by the congruence property. If $a^j \sim a^k$ for some $j < k < n$, the quotient has period $k - j < n$. If no n satisfies $a^n \sim \mathrm{id}$, all powers are distinct. □

Lemma 4.5 (Reduction to minimal period)**.** *Let (V, R, s, t) be an* invertible *relational system (with respect to a congruence $\sim$) with single generator a. If $a^n \sim \mathrm{id}$ for some $n \geq 1$ but the configurations $a^0, a^1, \ldots, a^{n-1}$ are not all pairwise inequivalent, then there exists a strictly smaller $k < n$ such that $a^k \sim \mathrm{id}$.*

Proof. If intermediate configurations are not all distinct, there exist $i < j \leq n$ with $a^i \sim a^j$. By the congruence property, composing both sides on the right with a^{-i} gives $a^i \circ a^{-i} \sim a^j \circ a^{-i}$, i.e., $\mathrm{id} \sim a^{j-i}$. Set $k := j - i < n$. Repeating if necessary yields a minimal period with all intermediate configurations pairwise inequivalent. □

Proposition 4.6 (Taxonomy of closure)**.** *Let (V, R, s, t) be a relational system with single generator a and congruence $\sim$, at its minimal period (all intermediate configurations distinct). Exactly one of the following holds:*

(i) ***Closed.*** *$a^n \sim \mathrm{id}$ for some $n \geq 1$. Quotient: $\mathbb{Z}/n\mathbb{Z}$.*

(ii) ***Pseudo-periodic.*** *$a^n \sim \phi$ for some $n \geq 1$ and some $\phi \sim a^j$ with $0 < j < n$ (so $\phi \not\sim \mathrm{id}$, but the orbit revisits a prior class). The system neither closes nor runs free.*

(iii) ***Open.*** *No periodicity law holds; all powers $a^0, a^1, \ldots$ are distinct. Quotient: $\mathbb{Z}$.*

Proof. By Lemma 4.5 it suffices to work at minimal period. Cases (i) and (ii) are distinguished by whether the return target is id or some $\phi \not\sim \mathrm{id}$. They cannot both hold simultaneously. Case (iii) is the negation of (i) and (ii). The three cases are mutually exclusive and exhaustive. □

4.2 Classification of Single-Generator Congruences

Definition 4.7 (Period and shift order)**.** Let (V, R, s, t) be a single-generator invertible relational system with generator a and congruence $\sim$.

(i) The *period* $n \in \mathbb{N} \cup \{\infty\}$ is the minimal $n > 0$ with $a^n \sim \mathrm{id}$, if it exists, and ∞ otherwise.

(ii) When $n < \infty$, define the *loop subgroup*

$$S := \left\{ [m] \in \mathbb{Z}/n\mathbb{Z} \;\middle|\; a^m \sim \mathrm{id} \right\} \subseteq \mathbb{Z}/n\mathbb{Z}.$$

The *shift order* is $k := n/|S|$.

Lemma 4.8. *The set S in Definition 4.7 is a subgroup of $\mathbb{Z}/n\mathbb{Z}$.*

Proof. If $a^m \sim \mathrm{id}$ and $a^p \sim \mathrm{id}$, then $a^{m+p} \sim \mathrm{id}$ by the congruence property. If $a^m \sim \mathrm{id}$, composing on the right by a^{-m} gives $\mathrm{id} \sim a^{-m}$. Hence S is closed under addition and negation in $\mathbb{Z}/n\mathbb{Z}$. □

Remark 4.9 (Interpretation of k). When $S = \{[0]\}$, no nontrivial power of a is identified with id: the system is a pure cycle and $k = n/1 = n$. The case $S = \mathbb{Z}/n\mathbb{Z}$ is unreachable at minimal period (Lemma 4.5). The relevant range is $1 < k \leq n$, with $k = n$ the closed case and $1 < k < n$ the pseudo-periodic cases. The shift order k equals the index $[\mathbb{Z}/n\mathbb{Z} : S]$.

Proposition 4.10 (Complete congruence classification)**.** *Let $R = \{a, a^{-1}\}$, and let $\sim_1$, $\sim_2$ be congruences on $\mathcal{C}$ with invariants (n_1, k_1) and (n_2, k_2). The quotients $\mathcal{C}/\sim_1$ and $\mathcal{C}/\sim_2$ are isomorphic if and only if one of the following holds:*

(i) $n_1 = n_2 = \infty$ (both open; both isomorphic to $\mathbb{Z}$);

(ii) $n_1 = n_2 < \infty$ and $k_1 = k_2 = n_1$ (both closed with the same period; $S_1 = S_2 = \{[0]\}$);

(iii) $n_1 = n_2 < \infty$ and $1 < k_1 = k_2 < n_1$ (both pseudo-periodic with the same period and shift order).

No other combination yields isomorphic quotients.

Proof. **Cases (i) and (ii).** Open congruences yield $\mathbb{Z}$; closed congruences with period n yield $\mathbb{Z}/n\mathbb{Z}$. These are isomorphic as monoids iff the periods agree. A finite monoid is never isomorphic to $\mathbb{Z}$.

Case (iii). *($\Rightarrow$)* If $\Phi : \mathcal{C}/\sim_1 \to \mathcal{C}/\sim_2$ is an isomorphism, it preserves period: $a^{n_1} \sim_1 \mathrm{id}$ implies $a^{n_1} \sim_2 \mathrm{id}$, giving $n_2 \leq n_1$; by symmetry $n_1 \leq n_2$, so $n_1 = n_2 =: n$. The isomorphism maps S_1 bijectively to S_2 (preserving identity and group operation), so $|S_1| = |S_2|$ and $k_1 = k_2$.

($\Leftarrow$) Assume $n_1 = n_2 = n$ and $k_1 = k_2$. Then $|S_1| = |S_2|$. Since $\mathbb{Z}/n\mathbb{Z}$ is cyclic it has exactly one subgroup of each order dividing n; therefore $S_1 = S_2 =: S$. Define $\Phi : \mathcal{C}/\sim_1 \to \mathcal{C}/\sim_2$ by $\Phi([a^j]_1) := [a^j]_2$.

Well-definedness. If $a^j \sim_1 a^{j'}$, then $a^{j-j'} \sim_1 \mathrm{id}$, so $[j - j'] \in S_1 = S_2$, hence $a^{j-j'} \sim_2 \mathrm{id}$ and $a^j \sim_2 a^{j'}$.

Bijectivity. $\Psi([a^j]_2) := [a^j]_1$ is the inverse by the same argument.

Composition-preservation. $\Phi([a^j]_1 \circ [a^m]_1) = [a^{j+m}]_2 = [a^j]_2 \circ [a^m]_2 = \Phi([a^j]_1) \circ \Phi([a^m]_1)$.

No other combinations. Closed quotients (finite, $k = n$) are not isomorphic to pseudo-periodic quotients (finite, $k < n$), nor to open quotients (infinite). □

Remark 4.11 (The (n, k) invariant). The pair (n, k) is a complete isomorphism invariant for single-generator quotients: it determines the quotient up to isomorphism and is determined by it. The parameter n measures orbit scale — how many steps before any return. The parameter k measures how far the system is from being purely cyclic: when $k = n$ the system is a pure cycle with no internal loop structure; as k decreases toward 1 the internal loop subgroup S grows, encoding increasingly rich repetition within each orbit. For multi-generator systems, the natural generalization is a normal subgroupoid of the free path groupoid; a complete invariant theory for that case remains open.

Remark 4.12 (Multi-generator classification gap). The framework is universal (Theorem 6.1): every finite metric space arises from some multi-generator relational system. Universality establishes expressiveness but not classification. The classification problem reduces to describing normal subgroupoids of free path groupoids up to isomorphism of quotient categories. Unlike the single-generator case, no complete invariant analogous to (n, k) is currently known, even for two generators with full invertibility. The difficulty is genuine: the corresponding word problem for finitely presented groupoids is in general undecidable, and the space of possible congruences is not well-understood even in simple cases.

5 Metric Emergence

Definition 5.1 (Weight Function)**.** A *weight function* is a map $w : R \to [0, \infty)$, extended to configurations by:

$$w(C) = w(r_1) + \cdots + w(r_n) \quad \text{for } C = (r_1, \ldots, r_n), \qquad w(\mathrm{id}) = 0.$$

When w takes values in $\mathbb{N}$, it is a *discrete weight function.*

Definition 5.2 (Relational Path Between Objects)**.** A *relational path from* $[A]$ *to* $[B]$ is a configuration $P \in \mathcal{C}$ such that for some representative $a \in [A]$, the configuration $a \circ P$ belongs to $[B]$. The set of all such paths is $\mathrm{Path}([A], [B])$.

Lemma 5.3 (Representative independence)**.** *The set* $\mathrm{Path}([A], [B])$ *and the value* $\inf\{w(P) \mid P \in \mathrm{Path}([A], [B])\}$ *are independent of the choice of representative $a \in [A]$.*

Proof. Let $a, a' \in [A]$, so $a \sim a'$. Suppose $P \in \mathcal{C}$ satisfies $a \circ P \in [B]$. Since $a \sim a'$ and $P \sim P$, the congruence property gives $a \circ P \sim a' \circ P$. Hence $a' \circ P \in [B]$, so $P \in \mathrm{Path}([A],[B])$ via a' as well. The set $\mathrm{Path}([A],[B])$ is therefore independent of representative. Independence of the infimum follows immediately, since the infimum is taken over the same set. □

Theorem 5.4 (Metric Emergence)**.** *Let (V, R, s, t) be an invertible relational system with congruence $\sim$ and weight function $w : R \to [0, \infty)$. Assume:*

(A1) $w(r) \geq 0$ *for all* $r \in R$,

(A2) $w(C_1) = w(C_2)$ *whenever* $C_1 \sim C_2$,

(A3) $w(\mathrm{id}) = 0$,

(A4) $w(r^{-1}) = w(r)$ *for all* $r \in R$.

Define:

$$d([A],[B]) = \inf\big\{w(P) \mid P \in \mathrm{Path}([A],[B])\big\}.$$

Then d is a pseudometric *on $\mathcal{C}/\!\sim$. If additionally:*

(A5) $d([A],[B]) = 0 \Rightarrow [A] = [B]$,

then d is a metric.

Proof. **Non-negativity.** Immediate from (A1).

Self-distance. $\mathrm{id} \in \mathrm{Path}([A],[A])$ and $w(\mathrm{id}) = 0$ by (A3), so $d([A],[A]) = 0$.

Symmetry. By invertibility and (A4), reversing $P = (r_1, \ldots, r_n)$ yields $P^{-1} = (r_n^{-1}, \ldots, r_1^{-1})$ from $[B]$ to $[A]$ with $w(P^{-1}) = w(P)$. Hence $d([A],[B]) = d([B],[A])$.

Triangle inequality. For $P \in \mathrm{Path}([A],[B])$ and $Q \in \mathrm{Path}([B],[C])$, the concatenation $P \circ Q \in \mathrm{Path}([A],[C])$ satisfies $w(P \circ Q) = w(P) + w(Q)$. Taking infima: $d([A],[C]) \leq d([A],[B]) + d([B],[C])$.

Metric. Assumption (A5) gives separation. □

Corollary 5.5 (Discrete metric emergence)**.** *If $w : R \to \mathbb{N}$ with $w(r) \geq 1$ for all $r \in R$ and R is finite, the infimum in Theorem 5.4 is attained:*

$$d([A],[B]) = \min\big\{w(P) \mid P \in \mathrm{Path}([A],[B])\big\}.$$

Proof. Any path of k steps has weight at least k. For fixed target weight M, the set of paths with $w(P) \leq M$ is finite (bounded length, finite R). The infimum over a finite non-empty set of non-negative integers is a minimum. □

Remark 5.6 (Separation condition). Assumption (A5) is not automatic. If $w(r) = 0$ for some $r \not\sim \mathrm{id}$, then $d([A],[A \circ r]) = 0$ even though $[A] \neq [A \circ r]$. The condition holds whenever $w(r) > 0$ for all $r \in R$.

6 Metric Universality

6.1 Reconstruction of Finite Metric Spaces

Theorem 6.1 (Reconstruction)**.** *Let (X, d) be a finite metric space. Then there exists a relational system (V, R, s, t) with weight function $w : R \to [0, \infty)$ such that the induced metric on $\mathcal{C}/\!\sim$ is isometric to (X, d).*

Proof. Set $V := X$, $R := \{e_{xy} \mid x, y \in X\}$, and define $s(e_{xy}) := x$, $t(e_{xy}) := y$, and $w(e_{xy}) := d(x, y)$. The congruence $\sim$ is generated by all equalities implied by the metric structure; the identity at x is $\mathrm{id}_x = e_{xx}$ with $w(e_{xx}) = 0$.

The single-edge path (e_{xy}) gives $d_{\mathrm{ind}}(x, y) \leq d(x, y)$. For any path $\gamma = (e_{x_0x_1}, \ldots, e_{x_{n-1}x_n})$, the triangle inequality for d gives $d(x, y) \leq \sum_i d(x_i, x_{i+1}) = w(\gamma)$. Taking the infimum: $d \leq d_{\mathrm{ind}}$. Hence $d_{\mathrm{ind}} = d$, and the identity map is an isometry. □

Remark 6.2 (Expressibility and canonicity). Theorem 6.1 establishes *expressibility*: every finite metric space encodes as a relational system. It does not establish canonicity: there is no unique encoding. Different relational presentations of the same point set, with different weight functions, correspond to different structural comparisons.

6.2 Minimal Presentations

Definition 6.3 (Redundant generator)**.** A generator $e_{xy} \in R$ is *redundant* if there exists a configuration $\gamma \in \mathrm{Conf}(x, y)$ not using e_{xy} with $w(\gamma) = d(x, y)$.

Theorem 6.4 (Minimal presentation)**.** *Every finite metric space (X, d) admits a generating relational system with no redundant generators, minimal with respect to edge inclusion.*

Proof. Start from the complete-graph construction of Theorem 6.1. Iteratively remove redundant generators; finiteness of X guarantees termination. Removing a redundant generator preserves the induced metric. No remaining generator can be removed without increasing some distance. □

Remark 6.5 (Non-uniqueness of minimal presentations)*.* Minimal presentations are not unique in general. If $d(x, z) = d(x, y) + d(y, z)$ for some intermediate point y, the edge e_{xz} is redundant and may be omitted; otherwise it is not. Non-uniqueness arises precisely where the triangle inequality is tight.

Proposition 6.6 (Minimal presentations and geodesic generators)**.** *A generator e_{xy} is non-redundant in a minimal presentation of (X, d) if and only if the direct step $x \to y$ is a geodesic segment — that is, no path through intermediate points achieves the same weight. Equivalently:*

$$e_{xy} \text{ is non-redundant} \iff d(x, y) < d(x, z) + d(z, y) \text{ for all } z \neq x, y.$$

The non-uniqueness of minimal presentations is governed exactly by the tight triangles of (X, d): for each triple with $d(x, z) = d(x, y) + d(y, z)$, the long edge e_{xz} may be included or omitted, yielding distinct but metrically equivalent minimal presentations.

Proof. A generator e_{xy} is redundant iff there exists an alternative path of equal weight — i.e., $d(x, y) = d(x, z) + d(z, y)$ for some intermediate z. The contrapositive gives non-redundancy iff no such z exists, which is the strict triangle inequality at every intermediate point. Non-uniqueness follows: whenever the triangle inequality is tight at some triple, the long-edge generator is optional. □

6.3 Domain of Validity

Theorem 6.7 (Domain of validity)**.** *The framework captures:*

(i) ***Finite metric spaces (exact)*** *via Theorem 6.1 and the categorical equivalence (Theorem 8.9).*

(ii) ***Compact metric spaces (GH-approximate)*** *via Theorem 9.2: any compact metric space is a Gromov–Hausdorff limit of finite relational systems.*

(iii) ***Locally finite non-compact spaces (pointed GH-approximate)*** *via Theorem 9.4.*

The following lie outside the current scope: non-locally-finite infinite spaces, smooth Riemannian structure (tangent spaces, differential forms), and continuous symmetry groups.

7 Examples

7.1 The Discrete Circle

Let $R = \{a\}$. Fix $n \geq 1$ and define the congruence generated by $a^n \sim \mathrm{id}$.

Verification. This is a congruence: if $k \equiv k' \pmod{n}$ and $\ell \equiv \ell' \pmod{n}$, then $k + \ell \equiv k' + \ell' \pmod{n}$.

Quotient. By Theorem 4.3:

$$\mathcal{C}/\!\sim\, = \{[a^0], [a^1], \ldots, [a^{n-1}]\} \cong \mathbb{Z}/n\mathbb{Z},$$

with $[a^k] \circ [a^\ell] = [a^{k+\ell \bmod n}]$.

Example 7.1 (Distance on the discrete circle)**.** Assign $w(a) = 1$. The distance between $[a^j]$ and $[a^k]$ is:

$$d\big([a^j], [a^k]\big) = \min\big(|k - j| \bmod n,\ n - (|k - j| \bmod n)\big).$$

The maximum distance is $\lfloor n/2 \rfloor$, reflecting circular topology. No coordinates or embedding are used.

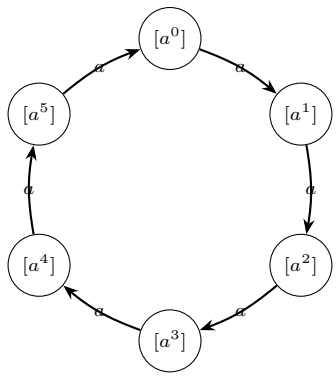

Figure 1: The discrete circle $\mathbb{Z}/6\mathbb{Z}$ as a relational quotient. The law $a^6 \sim \mathrm{id}$ forces closure.

7.2 End-to-End Worked Example: Four-Point Metric Space

Let $X = \{p, q, r, s\}$ with $d(p, q) = d(q, r) = d(r, s) = 1$, $d(p, r) = d(q, s) = 2$, $d(p, s) = 3$: four equally-spaced collinear points.

Starting from the complete-graph presentation (12 edges), the edges $e_{pr}, e_{rp}, e_{qs}, e_{sq}, e_{ps}, e_{sp}$ are all redundant (each has an alternative path of equal weight). The minimal presentation is the path graph $\{e_{pq}, e_{qp}, e_{qr}, e_{rq}, e_{rs}, e_{sr}\}$.

The induced metric recovers d exactly: $d_{\mathrm{ind}}(p, r) = 2$, $d_{\mathrm{ind}}(p, s) = 3$, etc. The triangle inequality is tight at every triple, which is precisely the condition for a unique minimal path presentation.

7.3 The Discrete Torus

Let $R = \{a, b\}$ with the congruence generated by:

$$a \circ b \sim b \circ a, \quad a^n \sim \mathrm{id}, \quad b^m \sim \mathrm{id}.$$

By Theorem 4.3:

$$\mathcal{C}/\!\sim \;\cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}.$$

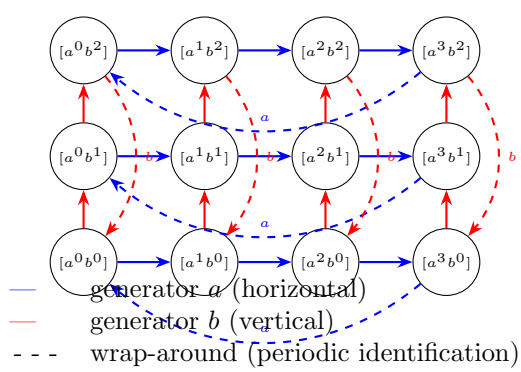

Figure 2: The discrete torus $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Dashed arrows indicate periodic identifications; commutativity $ab \sim ba$ means traversal order is irrelevant.

7.4 Relational Curvature via Noncommutativity

Let $R = \{a, b, a^{-1}, b^{-1}\}$ with $a \circ a^{-1} \sim \mathrm{id}$ and $b \circ b^{-1} \sim \mathrm{id}$. Define the *relational commutator*:

$$[a, b] = a \circ b \circ a^{-1} \circ b^{-1}.$$

If $[a, b] \sim \mathrm{id}$, the loop closes: the structure is locally flat. If $[a, b] \not\sim \mathrm{id}$, traversing the loop $a \circ b \circ a^{-1} \circ b^{-1}$ does not return to the starting class. This is the discrete relational analogue of holonomy: the signature of nonzero curvature.

Curvature is a discrete, finitely computable property: check whether $[a, b] \sim \mathrm{id}$ under the congruence.

Remark 7.2 (Algebraic vs. metric curvature). Nontrivial commutators detect *failure of local flatness* at the algebraic level: the loop $a \circ b \circ a^{-1} \circ b^{-1}$ does not return to its starting class, which is the discrete analogue of holonomy. However, noncommutativity alone does not imply negative metric curvature: free groups are highly noncommutative yet δ-hyperbolic; nilpotent groups are noncommutative but not hyperbolic. Metric curvature is captured quantitatively by the triangle defect κ of Proposition 7.3. The commutator $[a, b] \not\sim \mathrm{id}$ is a necessary algebraic precondition for curvature; the triangle defect provides the metric measurement. The two notions coincide — a nontrivial commutator produces a nonzero triangle defect — only under additional geometric conditions such as local regularity of the presentation.

Proposition 7.3 (Triangle defect and δ-hyperbolicity)**.** *Let $(\mathcal{O}, d)$ be the metric space induced by a relational system with inverses. Define the* triangle defect*:*

$$\kappa(x, y, z) := d(x, y) + d(y, z) - d(x, z).$$

(i) If $\kappa(x, y, z) \leq \delta$ for all $x, y, z \in \mathcal{O}$, then $(\mathcal{O}, d)$ is $O(\delta)$-hyperbolic in the sense of Gromov.

(ii) If $(\mathcal{O}, d)$ is δ'-hyperbolic, then $\kappa(x, y, z) \leq \delta'$ for all x, y, z.

In particular, $\kappa \equiv 0$ characterizes tree-like (0-hyperbolic) metric spaces.

Proof. (i). Assume $\kappa(x, y, z) \leq \delta$ for all triples, i.e., $d(x, z) \geq d(x, y) + d(y, z) - \delta$. Take any geodesic triangle with vertices x, y, z and a point p on side $[x, y]$ with $d(x, p) = a$, $d(p, y) = b$, $a + b = d(x, y)$. Applying the defect bound to (x, p, z) gives $d(p, z) \leq d(x, z) - a + \delta$. Comparing with the path $p \to y \to z$ of length $b + d(y, z)$ and using the defect bound for (x, y, z), a standard thin-triangle argument shows p lies within 2δ of $[x, z] \cup [y, z]$. Hence triangles are 2δ-thin and $(\mathcal{O}, d)$ is $O(\delta)$-hyperbolic. The precise constant relating defect bounds to the thin-triangle constant depends on the chosen definition of hyperbolicity; see [2] for a careful account.

(ii). If $(\mathcal{O}, d)$ is δ'-hyperbolic, apply the four-point condition with base point y: $d(x, z) + d(y, y) \leq d(x, y) + d(y, z) + \delta'$. Since $d(y, y) = 0$, this gives $\kappa(x, y, z) \leq \delta'$. □

7.5 Pseudo-Periodic Examples

Definition 7.4 (Pseudo-Periodic System)**.** A relational system with generator a and congruence $\sim$ is *pseudo-periodic* with period n and shift ϕ if $a^n \sim \phi$ and $\phi \not\sim \mathrm{id}$.

Example 7.5 (Helix)**.** Let $R = \{a, b\}$ with $a^n \sim b$ and $b^k \sim \mathrm{id}$. Each traversal of n steps in a advances one step in b; after k traversals the b-direction closes. The quotient encodes a discrete helix: n steps per turn, k turns before closure.

Remark 7.6 (Connection to semidirect products). If the shift ϕ acts on the cyclic structure by an automorphism φ, the resulting quotient is $\mathbb{Z}/n\mathbb{Z} \rtimes_\varphi \langle \phi \rangle$. The helix, screw motion, and glide reflection are all semidirect products of this form.

7.6 Negative Curvature: The Infinite Regular Tree

Fix $k \geq 2$. Let V be all finite words over $\{1, \ldots, k\}$; for each vertex v and symbol i, include $e_{v,i} : v \to vi$ and $\bar{e}_{v,i} : vi \to v$, with unit weights.

Proposition 7.7. *The induced metric satisfies $d(v, w) = |v| + |w| - 2|v \wedge w|$, where $v \wedge w$ is the longest common prefix.*

Theorem 7.8 (Hyperbolic features)**.** *The k-regular tree satisfies: (i) unique geodesics between any two vertices; (ii) exponential volume growth: k^n vertices at distance n from the root; (iii) 0-thin triangles: each side lies in the union of the other two.*

7.7 Positive Curvature: Triangulated Sphere

Let T be a triangulation of S^2 with vertex set V_T, edge set E_T, and unit weights. The induced metric d_T counts minimum triangulation edges in any path. The diameter is finite.

Example 7.9 (Geodesic non-uniqueness: icosahedral graph)**.** The icosahedral graph has 12 vertices, diameter 3, and $A_5 \times \mathbb{Z}/2\mathbb{Z}$ symmetry. Any vertex has exactly 6 shortest configurations of length 3 to its antipodal vertex: the relational analogue of multiple geodesics between antipodal points on a positively curved surface.

7.8 Flat Geometry: Integer Grid

Let $V_n = \{0, \ldots, n\}^2$ with all 8-connected edges of unit weight.

Proposition 7.10. *The induced metric is the Chebyshev distance:* $d_n((i_1, j_1), (i_2, j_2)) = \max\{|i_1 - i_2|, |j_1 - j_2|\}$.

The two generators a (horizontal) and b (vertical) commute, $a \circ b \sim b \circ a$, confirming $[a, b] \sim \mathrm{id}$: the structure is flat. The torus of Section 7.3 is this grid with periodic boundary conditions.

8 Categorical Equivalence

8.1 Lawvere Metric Spaces

Definition 8.1 (Lawvere metric space [1])**.** A *Lawvere metric space* is a set X with $d : X \times X \to [0, \infty]$ satisfying $d(x, x) = 0$ and $d(x, z) \leq d(x, y) + d(y, z)$. Symmetry is not required in general. A classical metric space is a symmetric Lawvere space with $d(x, y) = 0 \Rightarrow x = y$.

8.2 Relational Systems as Free Enriched Categories

Remark 8.2 (Generators and relations). In categorical language, a relational system (V, R, s, t) with congruence $\sim$ is a *category presented by generators and relations*: R supplies the generating morphisms and $\sim$ supplies the relations imposed on them. The word *relation* now carries its categorical meaning — a law imposed on composites — rather than its set-theoretic meaning. This is why the elements of R are called generators rather than relations throughout the paper.

Definition 8.3 (Free enriched category)**.** Given a weighted relational system (V, R, s, t, w), define a category enriched over $([0, \infty], \geq, +, 0)$ by: objects are states V; hom-value $\mathcal{C}(x, y) := \min_{\gamma \in \mathrm{Conf}(x,y)} w(\gamma)$ (∞ if no path exists). Composition: $\mathcal{C}(x, z) \leq \mathcal{C}(x, y) + \mathcal{C}(y, z)$.

Proposition 8.4. $(V, R, s, t, w) \mapsto \mathcal{C}$ *defines a well-formed* $([0, \infty], \geq, +, 0)$*-enriched category.*

Theorem 8.5 (Universal property)**.** *The assignment* $(V, R, s, t, w) \mapsto \mathcal{C}$ *is left adjoint to the forgetful functor from* $([0, \infty], \geq, +, 0)$*-enriched categories to weighted relational systems.*

Sketch. Given (V, R, s, t, w) and an enriched category $\mathcal{D}$, any weight-preserving map $\phi : R \to \mathcal{D}$ extends uniquely to an enriched functor $\mathcal{C} \to \mathcal{D}$ by sending each configuration to the composite of its images. □

8.3 Metric Realization Functor

Definition 8.6 (Category of relational systems)**.** **RelSys**: objects are weighted relational systems (V, R, s, t, w); morphisms $F : (V, R, s, t, w) \to (V', R', s', t', w')$ are maps $F_V : V \to V'$ and $F_R : R \to R'$ preserving source, target, and satisfying $w'(F_R(r)) \leq w(r)$ for all $r \in R$.

Theorem 8.7 (Metric realization functor)**.** *There is a functor* $\mathcal{M}$: **RelSys** $\to$ **LawMet** *defined by* $\mathcal{M}(V, R, s, t, w) := (V, d)$ *where* $d(x, y) = \mathcal{C}(x, y)$.

Remark 8.8 (Two metric levels and their relationship). The framework carries two distinct metric structures that must be distinguished.

The object metric (Theorem 5.4) is defined on $\mathcal{C}/\sim$: equivalence classes $[C]$ of configurations. For a single-state system ($|V| = 1$), all configurations are endomorphisms, $\mathcal{C}/\sim$ is a quotient monoid or group, and this metric is the word metric of that presentation. The examples of Section 7 (circle, torus, tree) all operate at this level.

The state metric (Definition above) is defined on V: the hom-value $\mathcal{C}(x, y)$ measures the minimum-weight path between states x and y. This is the quantity used in the reconstruction theorem and the categorical equivalence. It is the Lawvere metric structure on the underlying graph.

The bridge: in the canonical complete-graph presentation of Theorem 6.1, where $V = X$ and every pair of states has a direct generator, the state metric $\mathcal{C}(x, y)$ recovers the original metric $d(x, y)$ exactly. For multi-state systems the state metric is the appropriate geometric invariant; the object metric on $\mathcal{C}/\sim$ captures the richer process-equivalence structure within each hom-class. The categorical equivalence (Theorem 8.9) operates at the state metric level; an analogous equivalence at the object metric level would require tracking hom-sets, which is part of the framework's open program.

Theorem 8.9 (Equivalence of categories). *Let* $\mathbf{RelSys}_\sim$ *be* $\mathbf{RelSys}$ *with metrically equivalent objects identified. Then:*

$$\mathbf{RelSys}_\sim \simeq \mathbf{LawMet}_{\text{fin}}.$$

Proof. **Essential surjectivity.** By Theorem 6.1, every finite metric space arises as $\mathcal{M}(V, R, s, t, w)$.

Full faithfulness. *Injectivity.* If two morphisms F_1, F_2 induce the same map on states, they agree. *Surjectivity.* Given 1-Lipschitz $f : (V, d) \to (V', d')$, work in the canonical complete presentations: set $F_V = f$ and $F_R(e_{xy}) = e'_{f(x)f(y)}$. Then $w'(F_R(e_{xy})) = d'(f(x), f(y)) \leq d(x, y) = w(e_{xy})$ by 1-Lipschitz, so F is a morphism inducing f. □

9 Compatibility with Classical Geometry

Definition 9.1 (Metric approximation). A sequence of relational systems (G_n, w_n) with induced metrics (V_n, d_n) and maps $\phi_n : V_n \to X$ *approximates* a compact metric space (X, d) if: (i) $\phi_n(V_n)$ is ε_n-dense in X with $\varepsilon_n \to 0$; (ii) $|d_n(x, y) - d(\phi_n(x), \phi_n(y))| \leq \varepsilon_n$.

Theorem 9.2 (Gromov–Hausdorff bridge [2]). *If* (G_n, w_n) *approximates* (X, d)*, then* $d_{\text{GH}}((V_n, d_n), (X, d)) \to 0$.

Proof. Construct correspondences $\mathcal{R}_n$ from ϕ_n and a near-inverse ψ_n. Both maps satisfy $d(\phi_n(x), z) \leq \varepsilon_n$. The triangle inequality gives distortion $\leq 3\varepsilon_n$, hence $d_{\text{GH}} \leq \frac{3}{2}\varepsilon_n \to 0$. □

Corollary 9.3. *S^1, S^2, and $\mathbb{R}^2$ (locally) arise as Gromov–Hausdorff limits of finite relational systems via $\mathbb{Z}/n\mathbb{Z}$, geodesic triangulations, and integer grids respectively.*

Theorem 9.4 (Pointed GH emergence). *Let (G_n, w_n, v_n) be a sequence of rooted relational systems satisfying: (i) uniform local finiteness (degree $\leq D$); (ii) uniform edge bounds ($a \leq w_n(e) \leq b$); (iii) rooted local convergence (metric balls stabilize up to graph isomorphism for each fixed radius). Then (V_n, d_n, v_n) converges in the pointed Gromov–Hausdorff sense to a pointed metric space (V, d, v).*

Example 9.5 (The k-regular tree as pointed GH limit). Let G_n be the finite ball of radius n in T_k, with root v_0 and unit weights. The sequence satisfies all three hypotheses and converges to (T_k, d, v_0).

10 Multi-Object Configurations

Definition 10.1 (Inter-system relations). Let (G_A, w_A) and (G_B, w_B) be relational systems with disjoint vertex sets V_A, V_B. An *inter-system relational structure* is a weighted directed graph G_{AB} with edges in $V_A \times V_B \cup V_B \times V_A$ and weight function w_{AB}. The *combined relational system* is $(G_A \sqcup G_{AB} \sqcup G_B, w_A \sqcup w_{AB} \sqcup w_B)$.

Definition 10.2 (Combined metric)**.** The *combined metric* on $V_A \cup V_B$ is the induced metric of the combined system: $d(x, y) = \inf_\gamma w(\gamma)$ over all configurations γ in the combined system.

Theorem 10.3 (Space emergence)**.** *Let (G_A, w_A), (G_B, w_B) be finite relational systems and G_{AB} an inter-system structure. Then:*

(i) The combined system induces a finite metric space $(V_A \cup V_B, d)$ fully determined by G_A, G_B, and G_{AB}.

(ii) Different choices of G_{AB} over the same G_A, G_B produce genuinely different combined geometries.

Proof. (i) The combined system is a finite relational system; Theorem 5.4 applies. (ii) Different E_{AB} and w_{AB} produce different path sets, hence different infima. □

Special cases: $E_{AB} = \emptyset$ gives metrically disconnected union; a single bridge gives dumbbell geometry; zero-weight connections give metric gluing (relational pushout); uniform bipartite connections give product-like structure; position-varying weights give discrete warped products.

11 Relationship to Existing Work

The mathematical structures in this paper belong to a well-developed tradition. We state explicitly what is reformulated and what is new.

What is reformulated. A relational system (V, R, s, t) with congruence $\sim$ is a category presented by generators and relations in the standard sense [5, 6]. The quotient construction (Theorem 4.3) is the universal property of such a presentation. The metric emergence theorem (Theorem 5.4) is, in the single-state case, the word metric on a group presentation; in the multi-state case it is the path metric of the Cayley graph of the quotient groupoid. Both are classical objects in geometric group theory [7]. The categorical equivalence (Theorem 8.9) is a consequence of Lawvere's identification of metric spaces with categories enriched over $([0, \infty], \geq, +, 0)$ [1], combined with the reconstruction theorem. The Gromov–Hausdorff bridge results (Section 9) are standard applications of the GH framework [2].

What is new. The complete isomorphism classification of single-generator congruences by the invariant pair (n, k) (Proposition 4.10) reduces algebraically to the classification of subgroups of $\mathbb{Z}/n\mathbb{Z}$ by index, which is classical. The specific contribution is the formulation of this classification in relational-process language — characterizing period and shift order as intrinsic invariants of a congruence on path sequences — which does not appear in the existing literature on monoid and group presentations in this form. The pointed GH emergence theorem (Theorem 9.4) with its three explicit conditions is a concrete relational formulation not present in the metric geometry literature. The domain of validity theorem (Theorem 6.7), which organizes the boundary between exactly representable, GH-approximable, and out-of-scope spaces, is new as a unified statement.

The honest summary. The machinery is classical. The primary contribution is a presentation-level semantic framework: geometric structure is defined at the level of generators and congruence laws, prior to quotienting, and metric properties arise as invariants of congruence classes of processes. The framework isolates four layers — generators (syntax), congruence (laws), quotient (objects), metric (emergent distance) — that are usually conflated in geometric group theory expositions. The framework sits within the tradition of geometric group theory, graph geometry, and enriched category theory as one further development, not as a replacement for those foundations. The induced metric coincides with the path metric on the Cayley graph of the presented groupoid.

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